
Should we test two more kiosks and keep staffed lanes open?
Checkout
1
2
Now
Pilot option

What problem are we solving?

3 Hours (180 minutes) - Hour 1 (50 min): CoT concept lesson, prompt comparison, and live demonstration - Hour 2 (50 min): Guided assumption audit using a self-checkout case dataset - Hour 3 (50 min): Simple pilot review and improvement plan

SITUATION

The problem

How can the supermarket reduce its peak-hour queue time while preserving accessible customer service and controlling cost and risk? The store already has two self-checkout kiosks and staffed lanes. The current decision is whether to add two more kiosks for a 90-day pilot while keeping the staffed lanes open. The current average queue is 7.5 minutes during the busiest period. A one-line AI question can produce a confident recommendation without enough evidence about queue time, cost, shrinkage, or customer access.

TARGET

Learning goal

Use Chain-of-Thought (CoT) prompting as the solution method: define the problem, separate evidence from assumptions, compare alternatives, and test the most promising option before scaling it.

Practice material: CSV: session_1_4_self_checkout_pilot.csv

Use these supplied facts

Do not replace missing information with guesses.

CASE MATERIAL

Practice dataset

- Self-checkout pilot dataset (CSV) (data/session_1_4_self_checkout_pilot.csv). This is fictional instructional data created for this exercise. It is not a forecast or a real company record.

OPEN OR DOWNLOAD

Files for practice

- data/session_1_4_self_checkout_pilot.csv

Dataset rows 1-7 of 12

Dataset part 1 of 2. Use all parts as one practice dataset.

metric	period	value	unit	notes
peak_hour_transactions	weekday_17_00_to_19_00	84	transactions_per_hour	Instructional fictional case data
average_queue_time	weekday_17_00_to_19_00	7.5	minutes	Instructional fictional case data
cashier_hourly_wage	current	18	USD_per_hour	Instructional fictional case data
additional_kiosks	pilot	2	kiosks	Two additional kiosks while staffed lanes remain open
kiosk_monthly_lease	per_kiosk	1150	USD_per_month	Instructional fictional case data
monthly_support_cost	pilot	250	USD_per_month	Instructional fictional case data
baseline_inventory_shrinkage	current	1.2	percent_of_sales	Instructional fictional case data

Fictional instructional data from data/session_1_4_self_checkout_pilot.csv.

Dataset rows 8-12 of 12

Dataset part 2 of 2. Use all parts as one practice dataset.

metric	period	value	unit	notes
customers_age_65_plus	weekday_peak	20	percent_of_customers	Instructional fictional case data
pilot_duration	proposed	90	days	Instructional fictional case data
queue_time_target	pilot	5	minutes_or_less	Success threshold
shrinkage_ceiling	pilot	1.4	percent_of_sales	Stop rule threshold
kiosk_uptime_target	pilot	98	percent	Success threshold

Fictional instructional data from data/session_1_4_self_checkout_pilot.csv.

What students need to understand

Key explanations taken from the active session lesson.

CORE IDEA

What CoT is and why it is useful (15 min)

- CoT is a request for a structured decision rationale, not a request to reveal private model reasoning. In this course, students use it to make the business analysis inspectable.
- Define the decision: State the owner, deadline, alternatives, and success metric.
- Diagnose the evidence: Separate facts supplied in the case from assumptions and missing information.
- Compare alternatives: Assess each option using the same criteria, such as queue time, total operating cost, shrinkage risk, staff workload, and service access.
- Test before scaling: Give a conditional recommendation with a pilot target and a stop rule. CoT is useful because a business recommendation is only credible when a student can explain what evidence supports it and what new evidence would change it.

CORE IDEA

What happens without a structured rationale (10 min)

- A one-line question often forces the model to fill gaps with plausible but unverified details.
- False certainty: The response may give an exact profit forecast even though no traffic, cost, or loss data were supplied.
- Hidden assumptions: The response may assume customer acceptance, reliable technology, or adequate staff training without naming those assumptions.
- Unequal comparison: One option may receive detailed benefits while another option receives only a vague description.
- No operating plan: A recommendation may say "implement" without defining a pilot period, owner, target, or stop rule.

What students need to understand

Key explanations taken from the active session lesson.

CORE IDEA

The four-stage CoT decision frame (15 min)

- Define: What decision is being made? Who owns it? What outcome counts as success?
- Diagnose: Which facts are verified? Which assumptions must be tested? What information is missing?
- Compare: What are the realistic options? What are the benefits, trade-offs, and risks under the same criteria?
- Test: What 90-day pilot will provide evidence? What target means continue? What stop rule means pause or redesign?

CORE IDEA

Instructor live demonstration (10 min)

- Show the class the fictional self-checkout dataset. Ask students to predict which fields matter before they run a prompt.
- Known facts: 84 peak-hour transactions per hour, 7.5-minute average queue time, two proposed kiosks, $1,150 monthly lease per kiosk, $250 monthly support cost, 1.2% baseline shrinkage, and 20% of peak shoppers aged 65 or older.
- Pilot thresholds: Average queue time at or below five minutes, kiosk uptime at or above 98%, and shrinkage at or below 1.4% of sales.
- Discussion question: Which of these values is a fact, which is a target, and which value would you still need before approving a full rollout?

What makes a prompt weak?

These examples come from the current session scenario.

AVOID

Weak example 1

```
Should our grocery store add two more self-
checkout kiosks while keeping staffed lanes open?
```

AVOID

Weak example 2

```
We have 12 employees. Tell us exactly how much
profit two kiosks will add next year and
guarantee that customers will accept them.
```

A clearer, safer prompt

Keep the source, task, constraints, and output check visible.

USE

Improved example 1

```
You are helping a supermarket owner evaluate
adding two more self-checkout kiosks while
keeping staffed lanes open.

    Known facts:
    - 12 employees and two existing kiosks
    - Goal: reduce peak-hour queues
    - Concerns: theft, downtime, and service
access

    Create three labeled lists:
    1. Facts provided
    2. Assumptions that need validation
    3. Data needed before deciding

    Do not estimate costs or benefits that are
not in the facts.
    End with five questions for the owner.
```

USE

Improved example 2

```
You are a retail operations analyst.
    Use only the Session 1.4 self-checkout pilot
dataset.

    Compare these options:
    - Keep the current checkout model
    - Add two more kiosks for 90 days while
keeping staffed lanes open
    - Improve service without adding kiosks

    Evaluate every option using:
    - Customer wait time
    - Total operating cost
    - Shrinkage risk
    - Staff workload
    - Service access for older shoppers

    Recommend one option.
    State the assumptions that drive the
recommendation,
    one 90-day pilot target, and one stop rule.
```

Exercise 1: Follow the steps

Use the complete case and instructions from the session Markdown.

GUIDED PRACTICE

Exercise 1: Follow the steps

- Download the self-checkout pilot dataset (data/session_1_4_self_checkout_pilot.csv).
- Run Good Example 1 with the case facts.
- Make a three-column table: Fact, Assumption to Test, and Information Needed.
- Place at least three items in each column.
- Submit the table plus five questions you would ask the store owner.
- Success check: A fact comes from the dataset. An assumption is plausible but unverified. An information need is a value that would materially change the decision.

Exercise 2: Try a small variation

Use the complete case and instructions from the session Markdown.

APPLIED PRACTICE

Exercise 2: Try a small variation

- Scenario: The 90-day pilot has finished. The target queue time was five minutes or less. The actual average queue time was 6.5 minutes. Several customers said they needed help using the kiosks during the busiest hour.
- What this means: The pilot did not meet its queue-time target. This does not automatically mean that kiosks are a bad idea. It means the store should identify one practical change and test it before expanding the program.
- Step 1 - Name the problem: Write one clear sentence: "The peak-hour queue is 6.5 minutes, which is above the five-minute target."
- Step 2 - Choose one possible cause: Select one simple cause, such as unclear kiosk instructions, too few staff available to help, or customers needing more time to learn the system.
- Step 3 - Choose one improvement: Select one action, such as placing one employee near the kiosks during peak hours, adding a short instruction sign, or keeping one staffed lane open.
- Step 4 - Choose one metric: Measure the average peak-hour queue time again for two weeks. The target remains five minutes or less.
- Deliverable: A short three-bullet plan. Each bullet must state the problem, the action, and the number that will show whether the action worked.

Write your session report

Use Report_template.docx in Microsoft Word. Save it as FirstName_StudentID_W4_Keyword.docx and upload it to iLearn by 23:59 on class day.

SUBMIT

One report for this session

- Introduction: State the supermarket's queue problem and the decision the owner must make.
- Method: Copy the Chain-of-Thought (CoT) prompt you used in Exercise 1 and any prompt you used in Exercise 2.
- Results: Include the Fact, Assumption to Test, and Information Needed table with at least three items in each column, and your five questions for the store owner.
- Conclusion: Give your three-bullet plan from Exercise 2: the problem, the action, and the number that will show whether the action worked.

How to know the work is complete

Check your report before you hand it in.

VERIFY

Success check

- The answer follows the session constraints, uses supplied evidence, and labels missing information instead of guessing.